

Solve each system of equations using substitution for questions 1-4.

1.
$$\begin{cases} 3x + y = -9 \\ x = 6y \end{cases}$$

2.
$$\begin{cases} y = 2x \\ y = -2x + 12 \end{cases}$$

3.
$$\begin{cases} 2x + 3y = 5 \\ x = -2y \end{cases}$$

4.
$$\begin{cases} 4y - 1 = x \\ 8x + 7y = -47 \end{cases}$$

Use the system $\begin{cases} 3x - 2y = 11 \\ y = \frac{1}{2}x - 1 \end{cases}$ for questions 5 - 6.

5. Describe why using substitution is an efficient method to solving the system.

6. Solve the system.

John and Max work at a sandwich shop. John can make 15 sandwiches per hour, and Max can make 10 sandwiches per hour. Max worked 5 more hours than John and they made a total of 150 sandwiches that day. Determine the number of hours Max worked and the number of hours John worked.

7. Name the *two* variables that are being related in this situation.

8. Write a system of equations to represent the number of hours Max and John worked.

9. Solve the system of equations using substitution.

10. Interpret the solution of the solution.